# Detecting Heart Abnormality Using ECG Features with CART

**Team: Dhriti Kamani – PES1UG24AM085, Dhruv R – PES1UG24AM086**

## Abstract

We build a reproducible Python project that classifies patients of the UCI Arrhythmia data into heart-condition categories with two models: a pruned CART decision tree (the method of Zhou, Garcia and Rajbhandary, Stanford) and a Random Forest. Preprocessing is learned from training rows only, CART is tuned by cross-validation including cost-complexity pruning, and both models are evaluated on a held-out 30 % test set and by 10-fold CV on all rows. Random Forest reached 0.728 test accuracy (macro-F1 0.369) against 0.632 (0.237) for CART; always predicting class 1 scores 0.542. We do not reach the paper's ~80 %. The models use precomputed features; this is an academic exercise, not a diagnostic tool.

## 1. Problem statement

Given 279 recorded attributes of a patient (demographics and ECG-derived measurements), predict the class label (1 = normal; other labels = abnormal categories). This is multiclass and highly imbalanced.

## 2. Objectives

(i) validate and describe the dataset; (ii) preprocess without data leakage; (iii) tune and prune CART using training data only; (iv) compare it with a Random Forest; (v) evaluate on unseen data and identify influential features; (vi) compare honestly with the reference paper and state limitations.

## 3. Dataset

UCI Arrhythmia (archive.ics.uci.edu/dataset/5): 452 patient rows, 279 predictors (206 linear-valued, 73 nominal-coded) and one class column with labels from 1 to 16 (not every label occurs; class 1 = normal). The comma-separated file has no header and uses "?" for missing values. Rare classes have only a few rows, so accuracy alone is misleading.

## 4. Methodology

Load and validate → split 70/30 (stratified, seed 42) → inside one scikit-learn Pipeline: drop features >50 % missing, median-impute, fit model → CART: grid search over criterion, max_depth, min_samples_leaf and ccp_alpha (pruning) with 5-fold stratified CV on training rows → Random Forest (300 trees) → evaluate both once on the test rows and by 10-fold CV (3 repeats) on all rows. The test rows never influence imputation values, dropped features or hyper-parameters.

## 5. Preprocessing

A small transformer inside the pipeline drops any feature whose training missing share exceeds 50 % (the paper dropped feature 14, j_angle, 376/452 = 83.2 % missing, by hand; we decide from measured missingness, never from position) and fills remaining gaps with the training median. Because it sits inside the pipeline, cross-validation re-learns it on each training fold and the test rows are never used. The paper imputed in R; this is a deliberate, simpler difference.

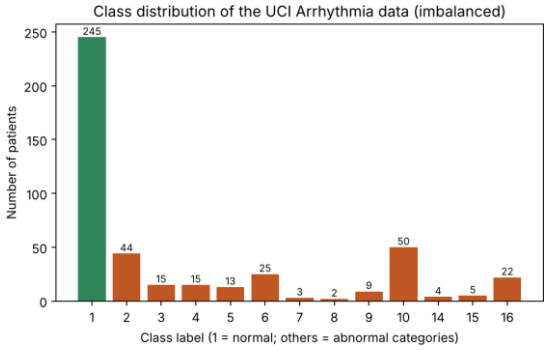



*Fig. A Class distribution (imbalanced)*

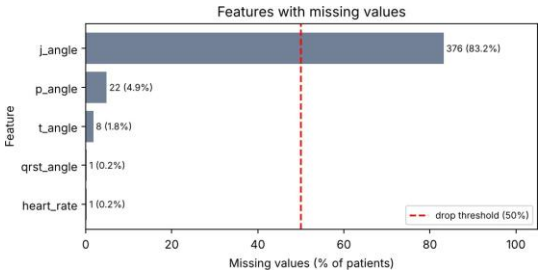



*Fig. B Features with missing values*

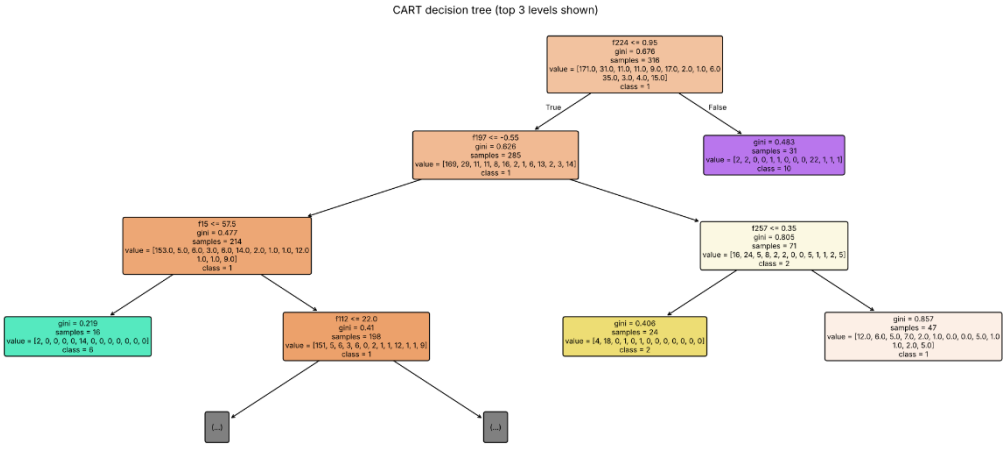



*Fig. C CART decision tree (top 3 levels, tuned and pruned)*

## 6. Implementation

Python 3, pandas, scikit-learn, Matplotlib. One script, main.py (plus an executed notebook ECG_CART_Analysis.ipynb), trains DecisionTreeClassifier (CART) and RandomForestClassifier and writes results.txt and three figures.

## 7. CART and pruning

CART greedily picks the feature/threshold that most reduces Gini impurity, recursively, giving an interpretable tree. A fully grown tree overfits, so ccp_alpha (cost-complexity pruning) is included in the CV grid {0, 0.005, 0.01, 0.02}. rpart's cp and scikit-learn's ccp_alpha are not numerically interchangeable, so the paper's 0.0001 / 0.027 are not reused. A Random Forest averages 300 such trees (bootstrap samples, random feature subsets): lower variance, less readable.

## 8. Experimental setup

316 train / 136 held-out test rows (seed 42); CART grid of 96 settings scored by 5-fold stratified CV on training rows only (selection metric = accuracy; training-CV 0.684). Selected: criterion=gini, max_depth=6, min_samples_leaf=3, ccp_alpha=0.02, giving 8 leaves. Random Forest: 300 trees, default settings. Dropped by the preprocessor: j_angle. Majority-class baseline: 0.542.

## 9. Results

On the held-out test set Random Forest reached 0.728 accuracy and 0.369 macro-F1; the pruned CART reached 0.632 and 0.237. Ten-fold CV on all rows (3 repeats, the paper's protocol) gives 0.739 and 0.669. The paper's ~80 % (R/rpart, its own protocol) is context only, not our result, and we did not reach it. The CV column uses a different protocol from the test columns, so they are not directly comparable; the RF-over-CART gap appears in both. In exploration, ExtraTrees and gradient boosting scored similarly to RF in training-CV (differences within noise); only CART and RF are in the shipped code.

| Metric (held-out test) | CART (pruned) | Random Forest |
|---|---|---|
| Accuracy | 0.632 | 0.728 |
| Balanced accuracy | 0.261 | 0.360 |
| Macro-F1 | 0.237 | 0.369 |
| 10-fold CV accuracy (all rows) | 0.669 | 0.739 |

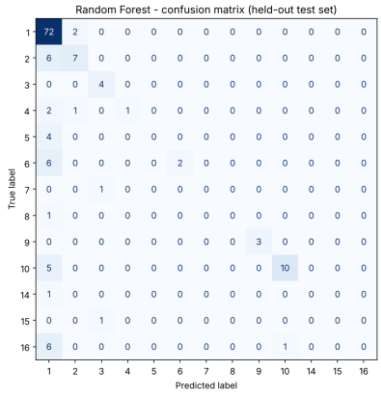



*Fig. 1 Confusion matrix (Random Forest, test)*

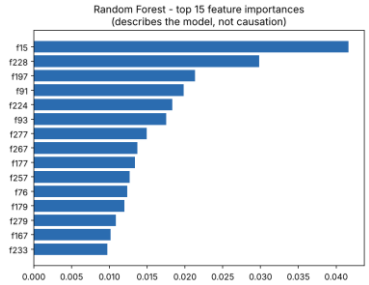



*Fig. 2 Random Forest feature importance*

## 10. Confusion-matrix observations

Random Forest: class 1 recall = 0.97 (72 of 74), but classes 5, 7, 8, 14, 15 and 16 are never recovered (recall 0) and class 4 and 6 recall is only 0.25. Macro-F1 (0.37) vs accuracy (0.73) shows how much the rare classes matter; balanced accuracy is 0.360 (CART 0.261).

## 11. Feature-importance observations

Random Forest top features: heart_rate (0.042), V1_qrsa (0.030), AVR_t_amp (0.021) (Fig. 2 labels them f15, f228, f197). The pruned CART roots at V1_rprime_amp (f224); the paper rooted at heart rate. Importance describes the model, not causation.

## 12. Conclusion

Random Forest improves clearly on a single pruned tree (test accuracy 0.728 vs 0.632; macro-F1 0.369 vs 0.237) at the cost of interpretability, but both stay well below the paper's ~80 % and fail on rare classes. We report what our own runs produced and did not tune toward a target figure.

## 13. Limitations and future work

Small data (452 rows, rare classes have only 1–8 test cases), a single 70/30 split (high variance), simple imputation, importance is not causal, no waveform input, no external validation. Future: nested CV, class weighting or resampling, better imputation, calibration, external data. **This project is an academic machine-learning experiment using a historical dataset. Its predictions are not medical diagnoses and must not be used for clinical decision-making. Clinical use would require rigorous independent validation, regulatory review, and appropriate medical oversight.**

## References

[1] B. Zhou, G. Garcia Jr, P. Rajbhandary, Detecting Heart Abnormality using ECG with CART, Stanford (course report). [2] UCI Machine Learning Repository, Arrhythmia data set, archive.ics.uci.edu/dataset/5/arrhythmia